

$^{36}\text{Ar}(\text{d}, ^3\text{He})$ 1974Do12

$J^\pi=0^+$ for ^{36}Ar ground state.

1974Do12: a 52-MeV deuteron beam was produced from the Karlsruhe isochronous cyclotron at the Max-Planck-Institut für Kernphysik. The target was 200-Torr 99.9% enriched ^{36}Ar gas. ^3He particles were detected using detector telescopes consisting of a 200- μm ΔE and a 2-mm E-surface-barrier counter with FWHM=125 keV. Measured $\sigma(E(^3\text{He}), \theta)$. Deduced levels, J, π , L-transfer, and spectroscopic factors from DWUCK-DWBA analysis.

Other: 1975Wa17 (theory discussions).

 ^{35}Cl Levels

Spectroscopic factor $C^2S=\sigma(\theta)_{\text{exp}}/\sigma(\theta)_{\text{DWBA}}/N$, where $N=2.95$ is a normalization factor used by 1974Do12 from 1966Ba54.

<u>E(level)[†]</u>	<u>Jπ[#]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>Jπ[#]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>	<u>E(level)[†]</u>	<u>Jπ[#]</u>	<u>L[‡]</u>	<u>C²S[‡]</u>
0	3/2 ⁺	2	2.2	3170 20	7/2 ⁻	3	0.22	6750 20	(5/2) ⁺	2	0.79
1220 20	1/2 ⁺	0	1.34	5150 20	(5/2) ⁺	2	0.09	6930 20	(5/2) ⁺	2	0.46
1763			<0.02	5600 20	(5/2) ⁺	2	0.47 19	8010 20	(5/2) ⁺	2	0.23
2700 20	3/2 ⁺	2	0.39	5720 20	(5/2) ⁺	2	0.73 29	8210 20	(5/2) ⁺	2	0.28
3000 20	5/2 ⁺	2	1.49	6140 20	(5/2) ⁺	2	0.72	8610 20	(5/2) ⁺	2	0.34

[†] From 1974Do12.

[‡] From DWBA analysis of the measured $\sigma(\theta)$ in 1974Do12.

[#] As given in 1974Do12 for levels below 5 MeV. For levels above 5 MeV, 1974Do12 states that they tend to interpret them as 5/2⁺ states because the summed low-lying $C^2S(1d_{3/2})$ already exceeds the simple shell model value for ^{36}Ar and the measured $\sigma(\theta)$ resembles more closely that of the 5/2⁺ state at 3.0 MeV than that of the 3/2⁺ ground state. All C^2S values are given according to the J values.